



PROBABILITY THEORY

MTM2592-Gr:2

SEVENTH WEEK

Assc. Prof. Dr. Ulku (BABUSCU) YESIL

ubabuscu@yildiz.edu.tr

ulkubabuscu@gmail.com

<https://avesis.yildiz.edu.tr/ubabuscu>

Example 1: Consider the caplets example.

- Find the conditional mean of X given Y=1.
- Find the conditional mean of Y given X=1.

		x			
		0	1	2	
y	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{12}$	$\frac{7}{12}$
	1	$\frac{2}{9}$	$\frac{1}{6}$		$\frac{7}{18}$
	2	$\frac{1}{36}$			$\frac{1}{36}$
		$\frac{5}{12}$	$\frac{1}{2}$	$\frac{1}{12}$	

Solution 1: a)

$$E(X|Y) = \sum_x xf(x|y) \quad E(X|Y=1) = \sum_{x=0}^2 xf(x|y=1)$$

$$f(x|y) = \frac{f(x,y)}{h(y)} = \begin{cases} f(0|1) = \frac{2/9}{7/18} = \frac{4}{7} \\ f(1|1) = \frac{1/6}{7/18} = \frac{3}{7} \\ f(2|1) = \frac{0}{7/18} = 0 \end{cases}$$

$$\begin{aligned} &= 0.f(0|1) + 1.f(1|1) + 2.f(2|1) \\ &= 0.\frac{4}{7} + 1.\frac{3}{7} + 2.0 = \frac{3}{7} \end{aligned}$$

b)

$$f(y|x) = \frac{f(x,y)}{g(x)} = \begin{cases} f(0|1) = \frac{1/3}{1/2} = \frac{2}{3} \\ f(1|1) = \frac{1/6}{1/2} = \frac{1}{3} \\ f(2|1) = \frac{0}{1/2} = 0 \end{cases}$$

$$E(Y|X=1) = \sum_{y=0}^2 yf(y|x=1)$$

$$\begin{aligned} &= 0.f(0|1) + 1.f(1|1) + 2.f(2|1) \\ &= 0.\frac{2}{3} + 1.\frac{1}{3} + 2.0 = \frac{1}{3} \end{aligned}$$

Example 2:

The joint probability function of X and Y is given below

$$f(x, y) = \begin{cases} k(x^2 - y), & 1 < x < 3, \quad 0 < y < 1, \\ 0, & \text{elsewhere} \end{cases}$$

- a) Find k,
- b) Find the joint distribution function (joint cumulative distribution) $F(X, Y)$,
- c) Find the marginal density of X and Y
- d) Compute $P(X < 2, Y < 0.5)$ and $P((X + Y) < 2)$.
- e) Are X and Y independent?
- f) $f(X | Y = 0.5) = ?$,
- g) $f(Y | X = 2) = ?$,
- h) Compute $P(X > 2 | Y > 0.5)$.
- i) Compute $P(Y < 0.4 | X > 2)$.
- j) Find the mean of X and Y random variables.
- k) Compute $\text{Cov}(X, Y)$,
- l) Compute $\text{Corr}(X, Y)$.

Solution 2:

a)

$$i) f(x, y) \geq 0 \quad ii) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = \int_0^1 \int_1^3 k(x^2 - y) dy dx = k \int_1^3 \left(x^2 y - \frac{y^2}{2} \right) \Big|_0^1 dx = k \int_1^3 \left(x^2 - \frac{1}{2} \right) dx = k \left[\frac{x^3}{3} - \frac{x}{2} \right]_1^3 = 1$$

$$\Rightarrow k \left[\frac{27}{3} - \frac{3}{2} - \left[\frac{1}{3} - \frac{1}{2} \right] \right] = k \left[\frac{23}{3} \right] = 1 \Rightarrow k = \frac{3}{23}$$

b) Joint distribution function (joint cumulative distribution)F(X,Y).

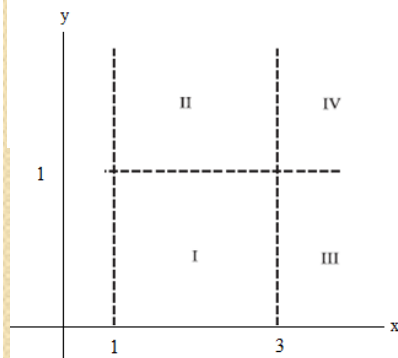
$$F(x, y) = \begin{cases} 0 & x \leq 1, y \leq 0 \\ F_1(x, y) & 1 < x < 3, 0 < y < 1 \\ F_2(x) & 1 < x < 3, y \geq 1 \\ F_3(y) & x \geq 3, 0 < y < 1 \\ 1 & x \geq 3, y \geq 1 \end{cases}$$

$$F(x, y) = P(X \leq x, Y \leq y) = \int_{-\infty}^y \int_{-\infty}^x f(s, t) ds dt \quad \text{for } -\infty < x < \infty, -\infty < y < \infty$$

$$\begin{aligned} F_1(x, y) &= P(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f(x, y) dy dx = \int_1^x \int_0^y \frac{3}{23} (u^2 - v) dv du = \frac{3}{23} \int_1^x \left(u^2 v - \frac{v^2}{2} \right) \Big|_0^y du \\ &= \frac{3}{23} \int_1^x \left(u^2 y - \frac{y^2}{2} \right) du = \frac{3}{23} \left[\frac{u^3 y}{3} - \frac{uy^2}{2} \right]_1^x = \frac{3}{23} \left[\frac{x^3 y}{3} - \frac{xy^2}{2} - \left(\frac{y}{3} - \frac{y^2}{2} \right) \right] = \frac{2x^3 y - 3xy^2 + 3y^2 - 2y}{46} \end{aligned}$$

$$\begin{aligned} F_2(x) &= P(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f(x, y) dy dx = \int_1^x \int_0^1 \frac{3}{23} (u^2 - v) dv du = \frac{3}{23} \int_1^x \left(u^2 v - \frac{v^2}{2} \right) \Big|_0^1 du \\ &= \frac{3}{23} \int_1^x \left(u^2 - \frac{1}{2} \right) du = \frac{3}{23} \left[\frac{u^3}{3} - \frac{u}{2} \right]_1^x = \frac{3}{23} \left[\frac{x^3}{3} - \frac{x}{2} - \left(\frac{1}{3} - \frac{1}{2} \right) \right] = \frac{2x^3 - 3x + 1}{46} \end{aligned}$$

$$\begin{aligned} F_3(y) &= P(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f(x, y) dy dx = \int_1^3 \int_0^y \frac{3}{23} (u^2 - v) dv du = \frac{3}{23} \int_1^3 \left(u^2 v - \frac{v^2}{2} \right) \Big|_0^y du \\ &= \frac{3}{23} \int_1^3 \left(u^2 y - \frac{y^2}{2} \right) du = \frac{3}{23} \left[\frac{u^3 y}{3} - \frac{uy^2}{2} \right]_1^3 = \frac{3}{23} \left[\frac{27y}{3} - \frac{3y^2}{2} - \left(\frac{y}{3} - \frac{y^2}{2} \right) \right] = \frac{52y - 6y^2}{46} \end{aligned}$$



$$F(x, y) = \begin{cases} 0 & x \leq 1, y \leq 0 \\ \frac{2x^3y - 3xy^2 + 3y^2 - 2y}{46} & 1 < x < 3, 0 < y < 1 \\ \frac{2x^3 - 3x + 1}{46} & 1 < x < 3, y \geq 1 \\ \frac{52y - 6y^2}{46} & x \geq 3, 0 < y < 1 \\ 1 & x \geq 3, y \geq 1 \end{cases}$$

c) Marginal density of X,

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy \quad \text{for } -\infty < x < \infty$$

$$g(x) = \int_0^1 \frac{3}{23} (x^2 - y) dy = \frac{3}{23} \left[x^2 y - \frac{y^2}{2} \right]_0^1 = \frac{3}{46} (2x^2 - 1)$$

$$\Rightarrow g(x) = \begin{cases} \frac{3}{46} (2x^2 - 1) & 1 \leq x \leq 3 \\ 0 & \text{elsewhere} \end{cases}$$

Marginal density of Y,

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx \quad \text{for } -\infty < y < \infty$$

$$h(y) = \int_1^3 \frac{3}{23} (x^2 - y) dx = \frac{3}{23} \left[\frac{x^3}{3} - xy \right]_1^3 = \frac{3}{23} \left[\frac{27}{3} - 3y - \left(\frac{1}{3} - y \right) \right]$$

$$= \frac{3}{23} \left[\frac{27 - 9y - 1 + 3y}{3} \right] = \frac{1}{23} [26 - 6y]$$

$$\Rightarrow h(y) = \begin{cases} \frac{1}{23} (26 - 6y) & 0 \leq y \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

d) $P(X < 2, Y < 0.5)$

$$P(X < 2, Y < 0.5) = \int_1^2 \int_0^{0.5} \frac{3}{23} (x^2 - y) dy dx = \frac{3}{23} \int_1^2 \left[x^2 y - \frac{y^2}{2} \right]_0^{0.5} dx$$

$$P(X < 2, Y < 0.5) = \frac{3}{23} \int_1^2 (0.5x^2 - 0.125) dx = \frac{3}{23} \left[\frac{0.5x^3}{3} - 0.125x \right]_1^2$$

$$P(X < 2, Y < 0.5) = 0.1685$$

$$P[(X+Y) < 2] = \int_1^2 \int_0^{2-x} \frac{3}{23} (x^2 - y) \, dy dx = \frac{3}{23} \int_1^2 \left[x^2 y - \frac{y^2}{2} \right]_0^{2-x} dx$$

$$\frac{3}{23} \int_1^2 \left[x^2(2-x) - \frac{(2-x)^2}{2} \right] dx = \frac{3}{23} \int_1^2 \left(2x^2 - x^3 - \left(\frac{4-4x+x^2}{2} \right) \right) dx$$

$$P[(X+Y) < 2] = \frac{3}{23} \left[\frac{2x^3}{3} - \frac{x^4}{4} - \frac{4x}{2} + x^2 - \frac{x^3}{6} \right]_1^2 \Rightarrow P[(X+Y) < 2] = 0,75$$

e) Are X and Y independent?

For independency $\forall (x, y) \rightarrow f(x, y) = g(x) \cdot h(y)$

$$f(x, y) = \begin{cases} \frac{3}{23}(x^2 - y) & 1 < x < 3 \quad 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases} \quad g(x) = \begin{cases} \frac{3}{46}(2x^2 - 1) & 1 \leq x \leq 3 \\ 0 & \text{elsewhere} \end{cases} \quad h(y) = \begin{cases} \frac{1}{23}(26 - 6y) & 0 \leq y \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

$$f(x, y) \stackrel{?}{=} g(x) \cdot h(y) \Rightarrow \frac{3}{23}(x^2 - y) \neq \frac{3}{46}(2x^2 - 1) \cdot \frac{1}{23}(26 - 6y)$$



X and Y random variables are dependent

$$\text{f) } P(X = x|Y = 0,5) = f(x|0,5) = \frac{f(x, 0,5)}{h(0,5)} = \frac{\frac{3}{23}\left(x^2 - \frac{1}{2}\right)}{\frac{1}{23}(26 - 6 \cdot (0,5))} = \frac{3}{46}(2x^2 - 1)$$

$$f(x|0,5) = \begin{cases} \frac{3}{46}(2x^2 - 1) & 1 < x < 3 \\ 0 & \text{elsewhere} \end{cases}$$

$$\text{g) } P(Y = y|X = 2) = f(y|2) = \frac{f(2,y)}{g(2)} = \frac{\frac{3}{23}(4-y)}{\frac{3}{46}(8-1)} = \frac{2}{7}(4-y)$$

$$f(y|2) = \begin{cases} \frac{2}{7}(4-y) & 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

h) $P(X > 2 | Y > 0,5)$

$$P(X > 2 | Y > 0,5) = \frac{P(X > 2, Y > 0,5)}{P(Y > 0,5)} = \frac{\int_2^3 \int_{0,5}^1 \frac{3}{23} (x^2 - y) dy dx}{\int_{0,5}^1 \frac{1}{23} (26 - 6y) dy}$$

$$\Rightarrow P(X > 2 | Y > 0,5) = 0,364$$

i) $P(Y < 0,4 | X > 2)$

$$P(Y < 0,4 | X > 2) = \frac{P(X > 2, Y < 0,4)}{P(X > 2)} = \frac{\int_2^3 \int_0^{0,4} \frac{3}{23} (x^2 - y) dy dx}{\int_2^3 \frac{3}{46} (2x^2 - 1) dx}$$

$$P(Y < 0,4 | X > 2) = \frac{\frac{3}{23} \int_2^3 \left(x^2 y - \frac{y^2}{2} \right) \Big|_0^{0,4} dx}{\frac{3}{46} \left| \frac{2x^3}{3} - x \right|_2^3} = \frac{\frac{3}{23} \int_2^3 (0,4x^2 - 0,08) dx}{\frac{35}{46}} = \frac{0,32}{0,76} = 0,42$$

j) E(X) and E(Y)

$$f(x, y) = \begin{cases} \frac{3}{23}(x^2 - y) & 1 < x < 3 \quad 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases} \quad g(x) = \begin{cases} \frac{3}{46}(2x^2 - 1) & 1 \leq x \leq 3 \\ 0 & \text{elsewhere} \end{cases} \quad h(y) = \begin{cases} \frac{1}{23}(26 - 6y) & 0 \leq y \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

$$\left. \begin{aligned} E(X) = \mu_x &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x \cdot f(x, y) dy dx = \int_{-\infty}^{\infty} x \cdot g(x) dx = \int_1^3 x \cdot \frac{3}{46}(2x^2 - 1) dx = \frac{48}{23} \\ E(Y) = \mu_y &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y \cdot f(x, y) dy dx = \int_{-\infty}^{\infty} y \cdot h(y) dy = \int_0^1 y \cdot \frac{1}{23}(26 - 6y) dy = \frac{11}{23} \\ E(XY) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy \cdot f(x, y) dy dx = \int_1^3 \int_0^1 xy \cdot \frac{3}{23}(x^2 - y) dy dx = \frac{26}{23} \end{aligned} \right\}$$

$$E(XY) \neq E(X)E(Y)$$

$$\frac{26}{23} \neq \frac{48}{23} \cdot \frac{11}{23} \quad \Rightarrow \quad \underline{\text{X and Y random variables are dependent}}$$

k) Cov(X,Y)

$$\sigma_{xy} = Cov(X, Y) = E(XY) - \mu_x \mu_y = \frac{26}{23} - \frac{48}{23} \frac{11}{23} = \frac{70}{529} = 0,1323$$

➡ There is a positive relationship between X and Y.

l) Corr(X,Y) = $\rho_{xy} = \frac{Cov(X, Y)}{\sqrt{Var(X) \cdot Var(Y)}} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$

$$Var(X) = E(X^2) - \mu_x^2 \quad Var(Y) = E(Y^2) - \mu_y^2$$

$$\mu_x = \frac{48}{23} \quad \mu_y = \frac{11}{23}$$

$$E(X^2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^2 \cdot f(x, y) dy dx = \int_{-\infty}^{\infty} x^2 \cdot g(x) dx = \int_1^3 x^2 \cdot \frac{3}{46} (2x^2 - 1) dx = \frac{661}{115}$$

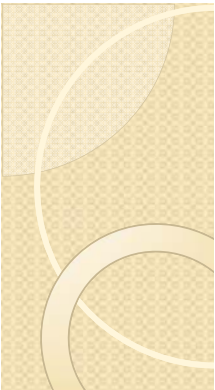
$$E(Y^2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y^2 \cdot f(x, y) dy dx = \int_{-\infty}^{\infty} y^2 \cdot h(y) dy = \int_0^1 y^2 \cdot \frac{1}{23} (26 - 6y) dy = \frac{43}{138}$$

$$Var(X) = \frac{661}{115} - \left(\frac{48}{23}\right)^2 = 1,3924$$

$$Var(Y) = \frac{43}{138} - \left(\frac{11}{23}\right)^2 = 0,0829$$

$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y} = \frac{0,1323}{\sqrt{(1,3924)(0,0829)}} = 0,3894$$

➡ There is a weak positive relationship between X and Y.



Example 3:

- For the traffic accidents happening in a highway, the random variables X represents the healthy situation of people and Y represents whether they wore a seat belt such as

$$X = \begin{cases} 0, & \text{dead} \\ 1, & \text{injured} \\ 2, & \text{healthy} \end{cases} \quad Y = \begin{cases} 0, & \text{wore a seat belt} \\ 1, & \text{did not wear a seat belt} \end{cases}$$

$X \backslash Y$	wore a seat belt	did not wear a seat belt
dead	45	230
injured	120	340
healthy	850	1015

According to the given table,

- find the joint probability function of X and Y ,
 - find the marginal distributions $g(x)$ and $h(y)$.
- 

Solution 3:

X\Y	wore a seat belt	did not wear a seat belt	Total
dead	45	230	275
injured	120	340	460
healthy	850	1015	1865
Total	1015	1585	2600

$$P(X = 0, Y = 0) = \frac{45}{2600} = 0.0173$$

$$P(X = 0, Y = 1) = \frac{230}{2600} = 0.088$$

$$P(X = 1, Y = 0) = \frac{120}{2600} = 0.0462$$

$$P(X = 1, Y = 1) = \frac{340}{2600} = 0.1308$$

$$P(X = 2, Y = 0) = \frac{850}{2600} = 0.3269$$

$$P(X = 2, Y = 1) = \frac{1015}{2600} = 0.3904$$

X\Y	0	1
0	0.0173	0.088
1	0.0462	0.1308
2	0.3269	0.3904

$$P(X = 0) = \sum_y P(x, y) = 0.0173 + 0.088 = 0.1053$$

$$P(X = 1) = \sum_y P(x, y) = 0.0462 + 0.1308 = 0.1770$$

$$P(X = 2) = \sum_y P(x, y) = 0.3904 + 0.3269 = 0.7173$$

$$P(X = 0) + P(X = 1) + P(X = 2) = 1$$

X\Y	0	1
0	0.0173	0.088
1	0.0462	0.1308
2	0.3269	0.3904

Marginal distribution of X

X	0	1	2
P(X=x)	0.1053	0.1770	0.7173

Marginal distribution of Y

$$P(Y = 0) = \sum_x P(x, y) = 0.0173 + 0.0462 + 0.3269 = 0.3904$$

$$P(Y = 1) = \sum_x P(x, y) = 0.088 + 0.1308 + 0.3904 = 0.6092$$

Y	0	1
P(Y=y)	0.6092	0.3904

$$P(Y = 0) + P(Y = 1) = 1$$

Example 4:

Let $f(x, y)$ be a joint probability function of X and Y . If $f(1,1) = 0.4$, $f(1,2) = 0.1$, $f(2,1) = 0.2$ and $f(2,2) = 0.3$, find the conditional probability of X given $Y = 1$.

Solution 4:

$$f(X|Y=1) = ?$$

$$f(Y=1) = \sum_x f(x,1) = f(1,1) + f(2,1) = 0.6$$

$$f(X=1|Y=1) = \frac{f(X=1, Y=1)}{f(Y=1)} = \frac{f(1,1)}{f(Y=1)} = \frac{0.4}{0.6} = \frac{4}{6} = \frac{2}{3}$$

$$f(X=2|Y=1) = \frac{f(X=2, Y=1)}{f(Y=1)} = \frac{f(2,1)}{f(Y=1)} = \frac{0.2}{0.6} = \frac{2}{6} = \frac{1}{3}$$

$$f(X|Y=1) = \begin{cases} 2/3, & X=1 \\ 1/3, & X=2 \\ 0, & \text{elsewhere} \end{cases}$$

Example 5:

A function is given as

$$f(x, y) = \begin{cases} \frac{1}{21}(2x - y), & x = 1, 2, 3, \quad y = 0, 1, \\ 0, & \text{otherwise} \end{cases}$$

- a) Is it a joint probability function?
- b) Find the marginal distributions X and Y .
- c) Compute $P(X \geq 2 \mid Y = 0)$.
- d) Compute $P(X \leq 2, Y = 2)$.
- e) Compute $P(X > Y)$.
- f) Are X and Y independent?

Solution 5:

a) i) $f(x_i, y_j) \geq 0$, for $\forall x_i, y_j$

ii) $\sum_{j=1}^{\infty} \sum_{i=1}^{\infty} f(x_i, y_j) = 1$

$$P(1, 0) = \frac{1}{21}(2 \cdot 1 - 0) = \frac{2}{21} > 0$$

$$P(1, 1) = \frac{1}{21}(2 \cdot 1 - 1) = \frac{1}{21} > 0$$

$$P(2, 0) = \frac{1}{21}(2 \cdot 2 - 0) = \frac{4}{21} > 0$$

$$P(2, 1) = \frac{1}{21}(2 \cdot 2 - 1) = \frac{3}{21} > 0$$

$$P(3, 0) = \frac{1}{21}(2 \cdot 3 - 0) = \frac{6}{21} > 0$$

$$P(3, 1) = \frac{1}{21}(2 \cdot 3 - 1) = \frac{5}{21} > 0$$

$$\sum_{x=1}^3 \sum_{y=0}^1 f(x, y) = f(1, 0) + f(1, 1) + f(2, 0) + f(2, 1) + f(3, 0) + f(3, 1) = 1$$

It is a joint probability function.

b) Marginal distribution of X

$$g(x) = \sum_{y=0}^1 f(x, y) = \frac{1}{21} [(2x - 0) + (2x - 1)] = \frac{1}{21} (4x - 1)$$

Marginal distribution of Y

$$h(y) = \sum_{x=1}^3 f(x, y) = \frac{1}{21} [(2.1 - y) + (2.2 - y) + (2.3 - y)] = \frac{1}{21} (12 - 3y)$$

$$c) \quad P(X \geq 2 | Y = 0) = \frac{P(X = 2, Y = 0) + P(X = 3, Y = 0)}{P(Y = 0)} = \frac{4/21 + 6/21}{12/21} = \frac{5}{6}$$

$$d) \quad P(X \leq 2 | Y = 2) = 0$$

$$e) \quad P(X > Y) = P(1, 0) + P(2, 0) + P(2, 1) + P(3, 0) + P(3, 1) = \frac{1}{21} [2 + 4 + 3 + 6 + 5] = \frac{20}{21}$$

$$f) \quad P(X, Y) \stackrel{?}{=} P(X)P(Y)$$

$$\frac{1}{21} (2x - y) \neq \frac{1}{21} (4x - 1) \cdot \frac{1}{21} (12 - 3y)$$

For example; for $x=1, y=0$

$$P(X, Y) = \frac{2}{21}, \quad P(X) = \frac{3}{21}, \quad P(Y) = \frac{12}{21}$$

So; X and Y are dependent.

Example 6:

The joint probability function of X and Y is given below.

X\Y	1	2	3	4	Sum
-1	0	0	0.1	0.2	0.3
0	0.1	0.1	0	0.1	0.3
1	0.2	0	0.1	0.1	0.4
Sum	0.3	0.1	0.2	0.4	1

- a) Find the marginal distribution of X random variable,
- b) Find the marginal distribution of Y random variable,
- c) Find $E(X)$ and $E(Y)$,
- d) Find $E(XY)$,
- e) Are X and Y independent?
- f) Find $E(3X+1)$ and $E(2X+4Y)$,
- g) Find $E(X^2+Y)$,
- h) Find $E(Y|X = 0)$,
- i) Find $Var(Y|X = 0)$.

Solution 6:

a) Marginal distribution of X

X	-1	0	1
P(X=x)	0.3	0.3	0.4

b) Marginal distribution of Y

Y	1	2	3	4
P(Y=y)	0.3	0.1	0.2	0.4

$$c) \quad E(X) = \sum_x xP(X) = (-1).(0.3) + (0).(0.3) + (1).(0.4) = 0.1$$

$$E(Y) = \sum_y yP(Y) = (1).(0.3) + (2).(0.1) + (3).(0.2) + (4).(0.4) = 2.7$$

$$d) \quad E(XY) = \sum_x \sum_y xyP(X, Y) = (-1).(1).(0) + (-1).(2).(0) + (-1).(3).(0.1) + (-1).(4).(0.2) + (0).(1).(0.1) + (0).(2).(0.1) + (0).(3).(0) + (0).(4).(0.1) + (1).(1).(0.2) + (1).(2).(0) + (1).(3).(0.1) + (1).(4).(0.1) = -0.2$$

e) $P(X, Y) \stackrel{?}{=} P(X)P(Y)$

For example; for $x=1, y=1$ $P(X=1, Y=1) = 0.2$, $P(X=1) = 0.4$, $P(Y=1) = 0.3$

$$0.2 \neq (0.4) \cdot (0.3)$$

$P(X=1, Y=1) \neq P(X=1)P(Y=1)$ So; X and Y are dependent.

f) $E(3X + 1) = 3E(X) + 1 = 3(0.1) + 1 = 1.3$

$$E(2X + 4Y) = 2E(X) + 4E(Y) = 2(0.1) + 4(2.7) = 11$$

g) $E(X^2 + Y) = E(X^2) + E(Y)$

$$E(X^2) = (-1)^2 \cdot (0.3) + (0)^2 \cdot (0.3) + (1)^2 \cdot (0.4) = 0.7$$

$$E(X^2 + Y) = E(X^2) + E(Y) = 0.7 + 2.7 = 3.4$$

$$h) \quad E(Y|X=0) = \sum y.P(Y|X=0)$$

$Y X = 0$	$P(Y X = 0) = \frac{P(Y, X = 0)}{P(X = 0)}$	$E(Y X = 0)$
1	0.1/0.3	1.(0.1/0.3)
2	0.1/0.3	2.(0.1/0.3)
3	0/0.3	3.(0/0.3)
4	0.1/0.3	4.(0.1/0.3)
Sum	1	0.7/0.3

$$i) \quad \text{Var}(Y|X=0) = E(Y^2|X=0) - [E(Y|X=0)]^2$$

$Y^2 X = 0$	$P(Y X = 0) = \frac{P(Y, X = 0)}{P(X = 0)}$	$E(Y^2 X = 0)$
1	0.1/0.3	1.(0.1/0.3)
4	0.1/0.3	4.(0.1/0.3)
9	0/0.3	9.(0/0.3)
16	0.1/0.3	16.(0.1/0.3)
Sum	1	2.1/0.3

$$\text{Var}(Y|X=0) = 2.1 / 0.3 - (0.7 / 0.3)^2 = 1.56$$

Example 7:

The probability function of X random variable, conditional distribution of Y given X=0 and X=2 respectively, given below.

$$P(X) = \begin{cases} 0.4, & x = 0 \\ 0.6, & x = 2 \\ 0, & \text{elsewhere} \end{cases}$$

$$P(Y|X = 0) = \begin{cases} 0.8, & y = 0 \\ 0.2, & y = 1 \\ 0, & \text{elsewhere} \end{cases}$$

$$P(Y|X = 2) = \begin{cases} 0.5, & y = 0 \\ 0.5, & y = 1 \\ 0, & \text{elsewhere} \end{cases}$$

- a) Find joint probability function of X and Y,
- b) Find $E(Y|X = 2)$,
- c) Find $P(X|Y = 0)$.
- d) Find $Var(X|Y = 0)$.

Solution 7:

$$P(Y|X) = \frac{P(X, Y)}{P(X)} \Rightarrow P(X, Y) = P(Y|X)P(X)$$

$$P(Y = 0|X = 0) = \frac{P(X = 0, Y = 0)}{P(X = 0)} \Rightarrow$$

$$P(X = 0, Y = 0) = P(Y = 0|X = 0)P(X = 0)$$

$$P(X = 0, Y = 0) = (0.8)(0.4) = 0.32$$

Other cells of the table are obtained similarly.

P(X,Y)	$y = 0$	$y = 1$
$x=0$	$(0.8)(0.4)$	$(0.2)(0.4)$
$x= 2$	$(0.5)(0.6)$	$(0.5)(0.6)$

P(X,Y)	$y = 0$	$y = 1$	Sum
$x=0$	0.32	0.08	0.40
$x= 2$	0.30	0.30	0.60
Sum	0.62	0.38	1.00

$$b) \quad E(Y|X = 2) = \sum_y y.P(Y|X = 2) = 0.(0.5) + 1.(0.5) = 0.5$$

$$c) \quad P(X|Y = 0) = \frac{P(X, Y = 0)}{P(Y = 0)} = \begin{cases} 0.32 / 0.62, & x = 0 \\ 0.30 / 0.62, & x = 2 \\ 0, & \text{elsewhere} \end{cases}$$

$$d) \quad \text{Var}(X|Y = 0) = E(X^2|Y = 0) - [E(X|Y = 0)]^2$$

$$E(X|Y = 0) = \sum_x xP(X|Y = 0) = 0.\left(\frac{0.32}{0.62}\right) + 2.\left(\frac{0.30}{0.62}\right) = \frac{30}{31}$$

$$E(X^2|Y = 0) = \sum_x x^2P(X|Y = 0) = 0^2.\left(\frac{0.32}{0.62}\right) + 2^2.\left(\frac{0.30}{0.62}\right) = \frac{60}{31}$$

$$\text{Var}(X|Y = 0) = \left(\frac{60}{31}\right) - \left(\frac{30}{31}\right)^2 = \frac{960}{961}$$

Example 8:

A function is given as

$$f(x, y) = \begin{cases} c(2x + 3y), & x = 0, 1, 2, \quad y = 0, 1, \\ 0, & \text{otherwise} \end{cases},$$

- a) If f is a probability function, find c
- b) Find $P(X)$ and $P(Y)$.
- c) Find $P(Y | X)$, $P(X | Y)$.
- d) Compute $P(Y > 0 | X \leq 1)$ and $P(X \leq 1 | Y > 0)$.

Solution 8:

$$\text{a) } \sum_{x=0}^2 \sum_{y=0}^1 c(2x + 3y) = 1$$

$$\sum_{x=0}^2 c(2x + 0) + c(2x + 3) = 1 \Rightarrow \sum_{x=0}^2 4cx + 3c = 1 \Rightarrow 21c = 1 \Rightarrow c = \frac{1}{21}$$

$$\text{b) } P(X) = \sum_{y=0}^1 \frac{1}{21} (2x + 3y) = \frac{2x}{21} + \frac{2x + 3}{21} = \frac{4x + 3}{21}$$

$$P(Y) = \sum_{x=0}^2 \frac{1}{21} (2x + 3y) = \frac{3y}{21} + \frac{2 + 3y}{21} + \frac{4 + 3y}{21} = \frac{9y + 6}{21}$$

$$c) \quad P(Y|X) = \frac{P(X, Y)}{P(X)} = \frac{\frac{1}{21}(2x+3y)}{\frac{4x+3}{21}} = \frac{2x+3y}{4x+3}$$

$$P(X|Y) = \frac{P(X, Y)}{P(Y)} = \frac{\frac{1}{21}(2x+3y)}{\frac{9y+6}{21}} = \frac{2x+3y}{9y+6}$$

$$d) \quad P(Y > 0 | X \leq 1) = \frac{P(0,1) + P(1,1)}{P(X=0) + P(X=1)} = \frac{\frac{1}{21} \cdot 3 + \frac{1}{21} \cdot 5}{\frac{3}{21} + \frac{7}{21}} = \frac{4}{5}$$

$$P(X \leq 1 | Y > 0) = \frac{P(0,1) + P(1,1)}{P(Y=1)} = \frac{\frac{8}{21}}{\frac{15}{21}} = \frac{8}{15}$$